



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Seventh Semester – Fall 2021

Paper: Operation Research

Course Code: STAT-412

Roll No.

Time: 3 Hrs. Marks: 60



Q.1. Answer the following short questions:

(6x5=30)

- Define operations research and discuss its role in decision making.
- Discuss the methods used to solve operation research models.
- Define linear programming and discuss its properties.
- Define graphical sensitivity analysis.
- Define Simplex method. Discuss its main feature and conditions.
- Discuss the iterative nature of the Simplex method.

Solve the following questions.

(3x10=30)

Q2. Solve the following problem using the graphical method.

subject to

$$\text{Maximize } Z = 80x_1 + 100x_2$$
$$\begin{aligned} x_1 + 2x_2 &\leq 720 \\ 5x_1 + 4x_2 &\leq 1800 \\ 3x_1 + x_2 &\leq 900 \end{aligned}$$

where

$$x_1, x_2 \geq 0$$

Q3. A carpenter has 90, 80 and 50 running feet respectively of teak, plywood and rosewood. The product A requires 2, 1 and 1 running feet of teak, plywood and rosewood respectively. While the product B requires 1, 2 and 1 running feet of teak, plywood and rosewood respectively. If A would sell for Rs. 480 and B would sell for Rs. 400 per unit, how much of each should be made and sell in order to obtain maximum gross income out of his stock of wood?

- Give a mathematical formulation to this linear programming problem.
- Use graphical method to solve the problem.
- Indicate clearly the feasible solution region on the graph.

Q4. Use Simplex method to solve the following problem. Provide the tables for all iterations.

subject to

$$\text{Maximize } Z = 3x_1 + 2x_2$$
$$\begin{aligned} 2x_1 + x_2 &\leq 40 \\ x_1 + x_2 &\leq 24 \\ 2x_1 + 3x_2 &\leq 60 \end{aligned}$$

where

$$x_1, x_2 \geq 0$$



UNIVERSITY OF THE PUNJAB

Seventh Semester 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Operation Research
Course Code: STAT-412

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Subjective Type

Q.2. Write short and concise answer to the following questions:

- Define OR and characteristics of OR
- Phases of solving a problem using OR
- Summarize the steps for solving a LP model
- Define the following with reference to LPP
 - Unbounded solution
 - Alternative optima
- How the problem of minimization can be solved by linear programming

(4+4+4+4+4)

Q.3. A jewelry store makes necklace and bracelets from gold and platinum. The store has 18 ounces of gold and 20 ounces of platinum. Each necklace requires 3 ounces of gold and 2 ounces of platinum, while each bracelet requires 2 ounces of gold and 4 ounces of platinum. The demand for bracelets is no more than four. A necklace earns \$300 in profit and a bracelet \$400. The store wants to determine the number of necklaces and bracelets to make in order to maximize profit.

- formulate a linear programming model for this problem
- solve it

(10)

Q.4. Provide the graphical solution to following problem

a) Min $Z = 4X_1 + X_2$

Subject to

$$3X_1 + X_2 = 3$$

$$4X_1 + 3X_2 \geq 6$$

$$X_1 + 2X_2 \leq 4$$

$$X_1, X_2 \geq 0$$

b) Maximize $Z = 80X_1 + 100X_2$ subject to

$$X_1 + 2X_2 \leq 720; \quad 5X_1 + 4X_2 \leq 1800; \quad 3X_1 + X_2 \leq 900; \quad X_1, X_2 \geq 0$$

(7+7)

Q.5. Provide simplex solution to the following problems

Minimize cost = $4X_1 + 5X_2$

Subject to $X_1 + 2X_2 \geq 2.7$

$$3X_1 + X_2 \geq 75$$

$$x_1 \geq 0, x_2 \geq 0$$

(6)



UNIVERSITY OF THE PUNJAB

Seventh Semester – 2019

Examination: B.S. 4 Years Program

Roll No.

PAPER: Operation Research

Course Code: STAT-412 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Write short and concise answer to the following questions:

(5 x 4 = 20)

- Define OR and briefly discuss its impact on society
- Phases of solving a problem using OR
- Explain the steps involved in setting up of a Simplex method
- Define the following with reference to L.P.P.
 - feasible solution
 - slack variable
- How it is ensured that variables are positive in graphical method.

Q.3. The cut-Right knife company sells sets of kitchen knives. The Basic Set consists of 2 utility knives and 1 chef's knife. The Regular Set consist of 2 utility knives and 1 chef's knife and 1 bread knife. The Deluxe Set consists of 3 utility knives, 1 chef's knife and 1 bread knife. Their profit is \$30 on Basic Set, \$40 on a Regular Set, and \$60 on a Deluxe Set. The factory has on hand 800 utility knives, 400 chef's knives and 200 bread knives. Assuming all sets are sold, how many of set should be sold to maximize the profit. What is the maximum profit?

(10)

Q.4. Provide the graphical solution to following problem

- Minimize $Z = 3X_1 + 8X_2$ subject to
 $X_1 + X_2 \geq 8$; $2X_1 - 3X_2 \leq 0$; $X_1 + 2X_2 \leq 30$; $3X_1 - X_2 \geq 0$; $x_1 \leq 10$; $x_2 \geq 9$; $X_1, X_2 \geq 0$
- Maximize $Z = 80X_1 + 100X_2$ subject to
 $X_1 + 2X_2 \leq 720$; $5X_1 + 4X_2 \leq 1800$; $3X_1 + X_2 \leq 900$; $X_1, X_2 \geq 0$

(7+7)

Q.5. Provide simplex solution to the following problems

Maximize $Z = 120X_1 + 40X_2 + 60X_3$
Subject to $X_1 + X_2 + X_3 \leq 100$
 $10X_1 + 4X_2 + 7X_3 \leq 500$
 $2X_1 + 3X_2 + X_3 \geq 60$ when $x_1, x_2, x_3 \geq 0$

(6)

PUACP



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2022

Paper: Operation Research

Course Code: STAT-412

Roll No. 032456

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

PART-1 (Short Essay Questions)

- Question1.** Answer briefly and to the point, short question carry for each marks: (4 x 5 = 20)
- Slack Variable
 - Constraints ✓
 - Optimal Solution
 - Degeneracy
 - Alternative Optima

- Question2.** A manufacturer of wooden articles produces chairs and tables which requires two types of input mainly wood & labour. The manufacturer knows that for a table three (3) unit of wood and One (1) unit of labour are required, while each for a chair 2 (two) unit of each. The profit from each table is Rs 20 and Rs 16 for each chair. The total available resources for manufacturer are 150 units wood and 75 units of labour. The manufacturer wants to maximize his profit. Formulate the problem mathematically. (10)

PART-2 (Long Essay Questions)

- Question3.** Provide graphical solution to following problem (8 + 7)

a. Maximize $z = 0.2x_1 + 0.3x_2$
Subject to:
 $3x_1 + 8x_2 \leq 576000$
 $5x_1 + 4x_2 \leq 480000$ ✓
 $5x_1 + 10x_2 \leq 500000$
 $x_1 \geq 30,000$
 $x_2 \geq 8,000$
 $x_1, x_2 \geq 0$

b. Minimize $z = 500x_1 + 100x_2$
Subject to:
 $90x_1 + 6x_2 \geq 160$
 $x + x_2 \geq 800$
 $x_1, x_2 \geq 0$

- Question4.** Provide simplex solution to following problem(at most three iterations) (8 + 7)

a. Maximize $z = 5x + 4y$
Subject to:
 $6x + 4y \leq 24$
 $x + 2y \leq 6$
 $-x + y \leq 1$
 $y \leq 2$
 $x, y \geq 0$

b. Minimize $z = 3x_1 + 6x_2 - 2x_3 + 4x_4$
Subject to:
 $x_1 + 7x_2 + 7x_4 \leq 46$
 $3x_1 - x_2 + x_3 + 2x_4 \leq 8$
 $2x_1 + 3x_2 - x_3 + x_4 \leq 10$
 $x_1, x_2, x_3, x_4 \geq 0$



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Seventh Semester – Fall 2024

Paper: Operation Research

Course Code: STAT-412

Roll No. [REDACTED]

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(5x6=30)

- Entering variable and Leaving variable
- Phases of solving a problem using Operational Research.
- Objective function.
- Model in Operational Research
- Unbounded and infeasible solution

Answer the following questions.

Q2. Each unit of food X costs Rs 25 and contains 2 units of protein and 4 units of iron while each unit of food Y costs Rs 30 and contains 3 units of protein and 2 units of iron. Each animal must receive daily at least 12 units of protein and 16 units of iron. How many units of each food should be fed to each animal to meet their protein and iron requirements with the smallest possible cost.

(10)

Q3. Solve the following problem using the graphical method.

(2x5=10)

(i) Maximize $Z = 3x_1 + 2x_2$

Subject to

$$2x_1 + x_2 \leq 100$$

$$x_1 + x_2 \leq 80$$

$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

(ii) Minimize $Z = 3X_1 + 8X_2$

Subject to

$$X_1 + 2X_2 \geq 30$$

$$2X_1 + X_2 \leq 6$$

$$X_1 \leq 10$$

$$X_1, X_2 \geq 0$$

Q4. Use Simplex method to solve the following problems. Provide the tables for all iteration.

(2x5=10)

(i) Maximize $Z = 2X_1 - 4X_2 + 5X_3 + 6X_4$

Subject to

$$X_1 + 4X_2 - 2X_3 + 8X_4 \leq 2$$

$$X_1 + 2X_2 + 3X_3 + 4X_4 \leq 1$$

$$X_1, X_2, X_3, X_4 \geq 0$$

(ii) Minimize $Z = 4x_1 + x_2$

Subject to

$$3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \leq 3$$

$$x_1 + 2x_2 \geq 4$$

$$x_1, x_2 \geq 0$$



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Seventh Semester 2018
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Operation Research
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TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

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Q.2 Answer briefly and to the point, each short question carry four marks. (20)

- Objective function
- Decision variable
- Uncontrollable variables.
- Constraints
- Deterministic model

Q.3 The cut-Right knife company sells sets of kitchen knives. The Basic Set consists of 2 utility knives and 1 chef's knife. The Regular Set consists of 2 utility knives and 1 chef's knife and 1 bread knife. The Deluxe Set consists of 3 utility knives, 1 chef's knife and 1 bread knife. Their profit is \$30 on Basic Set, \$40 on a Regular Set, and \$60 on a Deluxe Set. The factory has on hand 800 utility knives, 400 chef's knives and 200 bread knives. Assuming all sets are sold, how many of set should be sold to maximize the profit. What is the maximum profit? (10)

Q.4 Give a graphical solution for the following problem. (10)

$$\text{Maximize } Z = 80X_1 + 100X_2$$

$$\text{Subject to: } X_1 + 2X_2 \leq 720; \quad 5X_1 + 4X_2 \leq 1800; \quad 3X_1 + X_2 \leq 900; \quad X_1, X_2 \geq 0$$

Q.5 Provide simplex solution to following program. (10)

a) Minimize: cost = $4X_1 + 5X_2$
Subject to: $X_1 + 2X_2 \geq 2.7$
 $3X_1 + X_2 \geq 75$
 $x_1 \geq 0, x_2 \geq 0$

b) Maximize: $Z = 3X_1 + 2X_2$
With $X_1 + 2X_2 \leq 6$
 $2X_1 + X_2 \leq 8$
 $-X_1 + 2X_2 \leq 1$
 $X_2 \leq 2$
 $X_1, X_2 \geq 0$

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